Metro Ticket Booking Digitalization

Phase 2: Requirement Analysis

1. Introduction & Scope

The Requirement Analysis phase defines the specific functional and non-functional criteria the ServiceNow application must meet. This ensures the digital booking portal satisfies commuter needs while providing the transit administration with reliable backend control and reporting.

2. Functional Requirements

Req ID	Module	Description
FR-01	Service Portal	Commuters must be able to select an origin station, a destination station, and passenger count via a user-friendly form.
FR-02	Fare Engine	The system must auto-calculate the total journey fare by looking up distance values between the selected stations in a database.
FR-03	QR Generation	Upon payment/submission, a unique QR code string must be generated representing the ticket ID, validity, and stations.
FR-04	Admin Dashboard	Administrators require a backend view to add new stations, update fare matrices, and monitor live ticket volumes.

3. Non-Functional Requirements

- **Performance:** Ticket generation and QR code delivery must occur within 3 seconds of request submission to prevent commuter delays.
- **Availability:** The ticketing Service Portal must maintain 99.99% uptime, aligning with metro operating hours.
- **Responsiveness:** The Service Catalog item must be fully optimized for mobile browsers, as 90% of bookings are expected via smartphones.
- **Security:** Generated QR codes must be cryptographically signed or verified against the ServiceNow database at the turnstile to prevent forgery.
- **Scalability:** The Flow Designer architecture must handle up to 5,000 concurrent ticket requests during morning and evening transit peaks.

4. ServiceNow Platform Requirements

To deliver this solution, specific ServiceNow products and plugins must be activated and configured. The architecture relies on standard ITSM functionality paired with automation tools.

Service Catalog & Portal

Requires customized Record Producers or Catalog Items. The portal framework will host the commuter-facing widget, ensuring a clean, accessible layout.

Flow Designer

Replaces complex Business Rules for core logic. Will be required to orchestrate the fare calculation, record creation, and trigger external API calls.

IntegrationHub

Essential for connecting the instance to third-party QR code generation services or internal payment gateways securely via REST APIs.

System Notifications

Requires configured Email and SMS gateways (like Twilio integration) to dispatch the final digital ticket directly to the user's device.

5. Data & Integration Dependencies

External QR API Dependency: The system relies on a RESTful API to convert the generated Ticket ID and journey metadata into a PNG/SVG QR code image.

Turnstile Hardware Integration (Future Scope): While this phase covers generation, the generated ticket records in ServiceNow must eventually be exposed via a read-only API endpoint so that physical station turnstiles can validate the QR code upon scanning.